

Machine Learning and Structure Preservation with Applications

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**Workshop in industrial applications, numerical analysis
and machine learning**
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- Introduction
- REMODEL - MSCA Staff exchange project.
- Structure preservation for learning mechanics from data (collaboration with SINTEF).
- Viscoelastic networks (collaboration with TechnipFMC).
- Stability of neural networks (also in collaboration with PST).
- Improving Geometric Integrations with Deep Learning.
- Mixed precision Training of Neural ODEs.
- Norwegian Centre on AI for Decision Making.

Deep neural networks - function approximators

Let $\mathcal{V}$ input space, $\mathcal{W}$ output space

$$\varphi : \mathcal{V} \rightarrow \mathcal{W}.$$

DNNs - approximation of continuous functions:

$$\varphi \approx \varphi_{\theta}$$

by a composition of L simpler maps (layers)

$$\varphi_{\theta} = \varphi_L \circ \varphi_{L-1} \circ \cdots \circ \varphi_1, \quad \varphi_{\ell} = \varphi_{\theta_{\ell}} \quad \varphi_{\theta_{\ell}} : \mathcal{V}_{\ell-1} \rightarrow \mathcal{V}_{\ell}$$

$\mathcal{V}_0 = \mathcal{V}$ and $\mathcal{V}_L = \mathcal{W}$, each φ_{ℓ} depends on a finite number of parameters θ_{ℓ} .

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Example

$$\varphi_{\ell}(x) = \sigma(A_{\ell}x + b_{\ell}), \quad \theta_{\ell} = (A_{\ell}, b_{\ell})$$

composition of affine transformations and nonlinearities acting component-wise on vectors.

Deep neural networks - discretizing ODEs

Residual networks - numerical ODEs: $\mathcal{V}_{\ell-1} = \mathcal{V}_\ell = \mathcal{V}$ a compact subdomain of $\mathbb{R}^N$ and

$$\varphi_{\theta_\ell} = \text{id} + hf_{\theta_\ell}, \quad f_{\theta_\ell} : x \mapsto \sigma(A_\ell x + b_\ell), \quad \theta_\ell := (A_\ell, b_\ell)$$

can be seen as the forward Euler discretization of the flow map of the ODE

$$\dot{y} = \sigma(A(t)y(t) + b(t)), \quad y(0) = x, \quad t \in [0, h]$$

(He et al. 2015, Haber and Ruthotto 2017, and E 2017).

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ODE perspective

- Useful for designing new (*structure preserving*) neural networks.
- Useful for stability guarantees: 1-Lipschitz networks (WGANs and Learned Regularizers, Adversarial Robustness).
- Useful for learning vector fields.

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Deep Learning as optimal control: optimizing a cost function (distance) with respect to all the parameters

$$\min_{\varphi_\theta = \varphi_{\theta_L} \circ \dots \circ \varphi_{\theta_1}} E(\varphi_\theta) = \min_{\{\theta_\ell\}_{\ell=1}^L} E(\varphi_{\theta_L} \circ \dots \circ \varphi_{\theta_1})$$

is the **discretization of the optimal control problem:**

$$\inf_{\theta(t)} E(y(T)), \quad \text{subject to} \quad \dot{y} = f_\theta(t, y), \quad y(0) = x, \quad t \in [0, T],$$

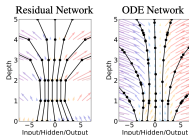
where $f_\theta(t, y(t)) = \sigma(A(t)y(t) + b(t)), \quad \theta(t) = (A(t), b(t)).$

Using the Pontryagin maximum principle we obtain the Hamiltonian BVP

$$\begin{aligned}\dot{y} &= f_{\theta}(t, y), \\ \dot{a} &= -\partial_y f_{\theta}(t, y)^T a \\ 0 &= \partial_{\theta} f_{\theta}(t, y)^T a\end{aligned}$$

with BCs $y(0) = x$, $a(T) = \nabla_y E|_{y(T)}$, one can construct NNs similar to ResNets.

- The solution of $\dot{y} = f_{\theta}(t, y)$ is obtained with an ODE solver (LSODE, VODE `scipy.integrate`)
- One option for backpropagation is solving the adjoint equation $\dot{a} = -\partial_y f_{\theta}(t, y)^T a$ backward in time both for a and recomputing y and with $y(T)$ and $a(T)$ as initial conditions.
- The output are solutions at $t = 0$ which are the gradients $\frac{\partial E}{\partial y}$ and $\frac{\partial E}{\partial \theta}$.



- Neural ODEs by Chen, Rubanova, Bettencourt, Duvenaud, *Neural Ordinary Differential Equations*, 2018
- Benning, EC, Ehrhardt, Owren, Schönlieb, *Deep learning as optimal control problems: models and numerical methods*, JCD, 2019

Normalizing flows

Problem: Given m observations of a random variable $x(t) \in \mathbb{R}^d$ estimate the underlying probability density $\rho(x)$.

Assume that ρ can be expressed as the composition of an invertible map φ^{-1} and a **known** probability density p e.g. the normal distribution.

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Change of variables $\varphi : \mathbb{R}^d \rightarrow \mathbb{R}^d$

$$y = \varphi^{-1}(x)$$

and the new probability density $p(y)$ so

$$\rho(x) = (\partial_y \varphi(y))^{-1} p(\varphi^{-1}(x))$$

We can approximate φ^{-1} with a Neural Network maximizing log-likelihood.

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$$\dot{z} = f(z(t), t), \quad z(0) = x, \quad y = z(1), \quad t \in [0, 1]$$

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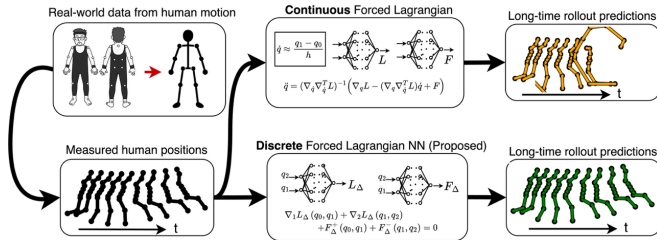
- **Strengths:** the computational advantage is that we deal with traces of Jacobians instead of determinants.
- **Weaknesses:** The approach can be hard to train and can be hard to scale to large datasets.
- **Fix:** Flow Matching.
 - Neural ODEs by Chen, Rubanova, Bettencourt, Duvenaud, *Neural Ordinary Differential Equations*, 2018
 - Y. Lipman, R. T. Q. Chen, H. Ben-Hamu, M. Nickel, and M. Le. *Flow Matching for Generative Modeling*, February 2023.

Research Exchanges in the Mathematics of Deep Learning with Applications is a **MSCA - Staff Exchange** project considering mathematical aspects of deep learning algorithms and their applications.



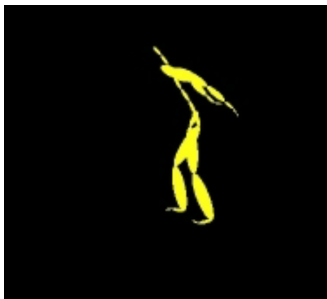
<https://www.se-remodel.eu/home>

Learning Mechanical systems from data



CMU Graphics Lab Motion Capture Database

- We assume to measure **only position data**.
- Necessary to take into account external forces and dissipative forces.
- We **use the laws of physics**: (discrete) Lagrange d'Alembert principle.
- We want to learn separately the **external forces** from the **conservative forces** (coming from the **Lagrangian**).
- M. Dyring Hansen, E. Celledoni, B. K. Tapley, *Learning mechanical systems from real-world data using discrete forced Lagrangian dynamics*, arXiv:2505.20370.



- Time curves on the manifold is $SO(3)^N$. Need to deal with Lie groups and Riemannian manifolds.
- We use stability of the geodesic Euler method to build neural networks with desired stability properties: 1-Lipschitz networks, α -averaged networks.
- M. Ghirardelli, B. Owren, E. Celledoni, *Conditional Stability of the Euler Method on Riemannian Manifolds*, arXiv:2503.09434.
- F Sherry, E Celledoni, MJ Ehrhardt, D Murari, B Owren, CB Schönlieb, *Designing stable neural networks using convex analysis and ODEs*, Physica D: Nonlinear Phenomena 463, 2024.

LdA principle

$$\delta \left(\int_{t_0}^{t_N} L(q(t), \dot{q}(t)) dt \right) + \int_{t_0}^{t_N} F(q(t), \dot{q}(t), u(t)) \cdot \delta q(t) dt = 0,$$

where δ denotes variations that vanish at the endpoints

$$q(t_0) = q_0 \text{ and } q(t_N) = q_N,$$

$L : TQ \rightarrow \mathbb{R}$ is the Lagrangian function

$F : TQ \rightarrow TQ^*$ is the Lagrangian force, a fiber preserving map

$$F : (q, \dot{q}) \mapsto (q, F(q, \dot{q})),$$

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Forced Euler- Lagrange Equations (equivalent to LdA)

$$0 = \mathcal{E}(L, F)(q, t) := \frac{\partial L}{\partial q}(q(t), \dot{q}(t)) - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}}(q(t), \dot{q}(t)) \right) + F(q(t), \dot{q}(t)).$$

LdA suggests that we should learn L separately from F .

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LdA suggests that we should learn L separately from F .

In the spirit of *Geometric Mechanics-Variational Integrators*, we use a **discrete LdA**.

- J E Marsden and M West, *Discrete variational integrators*, Acta Numerica, (2001);
- DM de Diego, R. Sato Martin de Almagro, *Variational order for forced Lagrangian systems*, Nonlinearity, 2018.

Discrete Lagrange d'Alembert principle: using only position data

$$L_{\Delta}(q_n, q_{n+1}) \approx \int_{t_n}^{t_{n+h}} L(q(t), \dot{q}(t)) dt,$$

$$F_{\Delta}^{-}(q_n, q_{n+1}, h) \cdot \delta q_n + F_{\Delta}^{+}(q_n, q_{n+1}, h) \cdot \delta q_{n+1} \approx \int_{t_n}^{t_{n+h}} F(q(t), \dot{q}(t)) \cdot \delta q(t) dt,$$

$L_{\Delta} : Q \times Q \rightarrow \mathbb{R}$ (discrete Lagrangian), $F_{\Delta}^{\pm} : Q \times Q \rightarrow T^*Q$ (discrete force)

$$\pi_Q^{-} = \pi_Q \circ F_{\Delta}^{-} \quad \pi_Q^{-} : Q \times Q \rightarrow Q, \quad \pi_Q^{+} = \pi_Q \circ F_{\Delta}^{+} \quad \pi_Q^{+} : Q \times Q \rightarrow Q,$$

with $\pi_Q : T^*Q \rightarrow Q$ the projection.

The discrete Lagrange-d'Alembert principle: find a discrete trajectory $\{q_n\}_{n=1}^N$ s.t.

$$\delta \sum_{n=0}^{N-1} L_{\Delta}(q_n, q_{n+1}) + \sum_{n=0}^{N-1} [F_{\Delta}^{-}(q_n, q_{n+1}) \cdot \delta q_n + F_{\Delta}^{+}(q_n, q_{n+1}) \cdot \delta q_{n+1}] = 0,$$

for all variations $\{\delta q_n\}_{n=0}^N$ vanishing at the endpoints $\delta q_0 = \delta q_N = 0$.

Discrete forced Euler-Lagrange equations:

$$0 = \mathcal{E}_{\Delta}(L_{\Delta}, F_{\Delta})(q_{n-1}, q_n, q_{n+1}) := D_2 L_{\Delta}(q_{n-1}, q_n) + D_1 L_{\Delta}(q_n, q_{n+1}) \\ + F_{\Delta}^{+}(q_{n-1}, q_n) + F_{\Delta}^{-}(q_n, q_{n+1}), \quad n = 2, \dots, N-1,$$

where D_1 and D_2 denote differentiation with respect to the first and second variable.

Goal: learn NNs $L_\theta \approx L$ and $F_\theta \approx F$ from the observed trajectories of the mechanical system

Inverse discrete LdA setting:

Given multiple observed (discrete) trajectories $\{q_n\}_{n=0}^N$ we seek approximations to the Lagrangian and forces (L, F) .

We replace (L, F) with neural networks approximations

$$L_\theta \approx L, \quad F_\theta \approx F$$

(L_θ, F_θ) are then the unknowns of our problem which we find minimizing the loss function

$$\mathcal{L} = \mathcal{L}_{\text{mech}}(L_\theta, F_\theta) + \mathcal{L}_{\text{reg}}$$

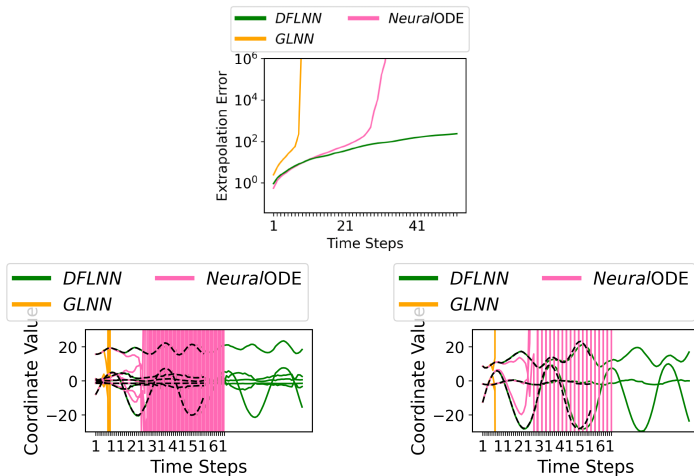
$\mathcal{L}_{\text{mech}}$: residual of discrete Euler Lagrange equations.

$\mathcal{L}_{\text{reg}}$: promotes that L_θ is a regular Lagrangian i.e.

$$S := \left(\frac{\partial^2 L_\theta(q, \dot{q})}{\partial \dot{q}_i \partial \dot{q}_j} \right),$$

is invertible.

Experiments (real-world data): swinging motion



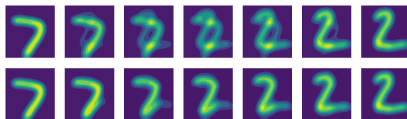
Reconstruction of human motion, capturing a swinging motion. (Black - recorded motion)

Top Extrapolation error.

Left Prediction rollouts in coordinate space for the root of the skeleton.

Right Prediction rollouts for the the right femur.

MNIST, matching of images, handwritten digits



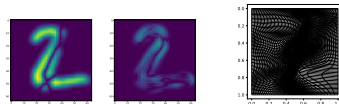
Top $\lambda(f_1, f_2, \tau) := \tau f_1 + (1 - \tau) f_2$

Bottom $\gamma(f_1, f_2, \tau) := \tau f_1 + (1 - \tau) f_2 \circ \varphi^*$, φ^* optimal reparametrization.

Optimal reparametrization problem:

$$d_S([c_0], [c_1]) = \inf_{\varphi \in \text{Diff}^+(I)} d_{\mathcal{P}}(c_0, c_1 \circ \varphi)$$

$$\varphi_\theta = (\text{id} + h_1 f_{\theta_1}) \circ \dots \circ (\text{id} + h_L f_{\theta_L}).$$



E. C., H. Glöckner, J. Riseth and A. Schmeding, *Deep learning of diffeomorphisms for optimal reparametrizations of shapes*, BIT, 2023.

Reformulation of the optimal reparametrization problem

Optimal reparametrization problem: given curves or surfaces c_0 and c_1 in $\mathcal{P}$ and shapes $[c_0]$ and $[c_1]$ in $\mathcal{S}$

$$d_{\mathcal{S}}([c_0], [c_1]) = \inf_{\varphi \in \text{Diff}^+(I)} d_{\mathcal{P}}(c_0, c_1 \circ \varphi)$$

Optimisation problem on an infinite dimensional Lie group $\text{Diff}^+(I)$, with "Lie algebra" $T_{\text{id}}\text{Diff}^+(I)$, $I = [0, 1]$ or $I = [0, 1]^2$.

To approximate we replace $\text{Diff}^+(I)$ with a finite dimensional space.

Consider a basis $v_1, v_2, \dots$ of $T_{\text{id}}\text{Diff}^+(I)$ write

$$\varphi_{\theta} = (\text{id} + h_1 f_{\theta_1}) \circ \dots \circ (\text{id} + h_L f_{\theta_L}).$$

$$f_{\theta_{\ell}} = \sum_{j=1}^M \beta_j^{\ell} v_j, \quad \theta_{\ell} = \{\beta_j^{\ell}\}_{j=1, \dots, M}^{\ell=1, \dots, L}$$

Optimize on these "approximate diffeomorphisms".

Viscoelastic properties of a multibody structure for fatigue life estimation of subsea equipment.

Sensors:

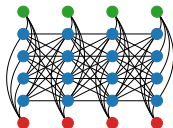
- Gyroscopes measure angular velocities and accelerometers measuring linear accelerations.
- Deep Water Sensor measuring strains.
- H. S. Gustad, E.C. and P.T. Moe, *Construction and analysis of viscoelastic networks for multibody systems*, in preparation.

Viscoelastic Networks

We consider the "Neural ODE" approach to ResNets:

- Design a suitable ODE, continuous counterpart of the network.
- Discretize with Forward Euler or another time discretization to obtain the NN.
- Include *desired physical properties* in the ODE. Discretize *preserving these properties*.

Our Networks are RNNs



- input: sensor signals
- output: bending moments

Based on simple **linear models for viscoelastic materials** (Maxwell model, Kelvin-Voigt, standard linear solid, Wiechert) and a **free energy approach satisfying the second law of thermodynamics**, we derived the following simple model equation

$$\begin{aligned}\dot{x}(t) &= (B(x) - A(x))\nabla\psi(x) + Du(t) \\ y &= D\nabla\psi(x)\end{aligned}$$

where $B(x)$ is skew-symmetric and $A(x)$ is SPD, this is an input-state-output port-Hamiltonian system. This system is input-output passive

$$\frac{d}{dt}\psi(x) = \nabla\psi^T B(x)\nabla\psi(x) - \nabla\psi(x)^T A(x)\nabla\psi(x) + \nabla\psi(x)^T Du \leq y^T u$$

Compatible with **second law of thermodynamics**.

- A van der Schaft and D Jeltsema, *Limits to Energy Conversion*, IEEE Trans Aut. Cont., 2022.

Discretizing the ODE: viscoelastic networks, map from layer to layer

- **Forward Euler:**

$$x_{n+1} = x_n + h (B(x_n) - A(x_n)) \nabla \psi(x_n) + u(t_n),$$

- **Averaged Vector Field method (AVF):**

$$x_{n+1} = x_n + h \left(B(x_{n+\frac{1}{2}}) - A(x_{n+\frac{1}{2}}) \right) \bar{\nabla} \psi(x_n, x_{n+1}) + u(t_n), \quad x_{n+\frac{1}{2}} = \frac{1}{2}(x_n + x_{n+1})$$

with input $u(t_n)$ and output $y_n = \bar{\nabla} \psi(x_n, x_{n+1})$ at time t_n .

Here $\bar{\nabla} \psi : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ is a **discrete gradient**, i.e. and approximation of the gradient with the properties:

$$\bar{\nabla} \psi(x, x) = \nabla \psi(x) \text{ and} \\ \psi(x') - \psi(x) = (x' - x)^T \bar{\nabla} \psi(x, x').$$

Using the properties of discrete gradients, for AVF we have

$$\begin{aligned} \psi(x_{n+1}) - \psi(x_n) &= \bar{\nabla} \psi(x_n, x_{n+1})^T (x_{n+1} - x_n) = \\ &\quad - h \bar{\nabla} \psi(x_n, x_{n+1})^T A(x_{n+\frac{1}{2}}) \bar{\nabla} \psi(x_n, x_{n+1}) + \bar{\nabla} \psi(x_n, x_{n+1})^T u(t_n) \\ &\leq \bar{\nabla} \psi(x_n, x_{n+1})^T u(t_n) \end{aligned}$$

Discrete passivity (compatibility with the second law of thermodynamics).

Conditional stability of ResNets (Forward Euler) is useful

- for Viscoelastic Networks, which are inherently passive and also Input-to-state stable (the solution is bounded).
- for adversarial robustness of ResNets.

Stability of ResNets - Contractivity of the underlying ODE

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A vector field $f(t, y)$ is **contractive** in ℓ^2 -norm if there is $\nu > 0$ such that for all y_1, y_2 and $t \in [0, T]$:

$$\langle f(t, y_2) - f(t, y_1), y_2 - y_1 \rangle \leq -\nu \|y_2 - y_1\|^2.$$

This implies that for any two integral curves $y(t)$ and $z(t)$

$$\|y(t) - z(t)\| \leq e^{-t\nu} \|y(0) - z(0)\|.$$

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The vector field

$$f(t, y(t)) = -A(t)^T \sigma(A(t)y(t) + b(t)),$$

with σ increasing function, $A \in \mathbb{R}^{n \times k}$, $b \in \mathbb{R}^n$, is **contractive**.

- EC, Ehrhardt, Etmann, McLachlan, Owren, Schönlieb, Sherry, *Structure preserving deep learning*, EJAM, 2021.

Theorem (Dahlquist and Jeltsch, 1979 - Theory of circle contractivity)

Suppose X satisfies the **cocoercivity condition**

$$\langle X(t, y_2) - X(t, y_1), y_2 - y_1 \rangle \leq -\bar{\nu} \|X(t, y_2) - X(t, y_1)\|^2, \quad \bar{\nu} \geq 0.$$

Then, if the stepsize h satisfies

$$h \leq 2\bar{\nu},$$

the **forward Euler method is non-expansive**.

Contractivity of forward Euler

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Proposition

For σ non decreasing and L -Lipschitz, the vector field

$$X(t, y) = -A(t)^T \sigma(A(t)y + b),$$

with $A \in \mathbb{R}^{n \times k}$, $b \in \mathbb{R}^n$, satisfies the cocoercivity condition with $\bar{\nu} = \frac{1}{\|A\|^2 L}$.

Remark X is a gradient vector field:

$$\dot{y} = -\nabla_y V, \quad V(t, y(t)) = \langle \gamma(A(t)y(t) + b(t)), \mathbb{1} \rangle, \quad \gamma' = \sigma.$$

- Sherry, EC, Ehrhardt, Murari, Owren and Schönlieb, *Designing Stable Neural Networks using Convex Analysis and ODEs*, 2023, arXiv:2306.17332, Physica D, 2024.

- Databases of images: using convolution neural networks
- Network:

$$\Psi = \varphi_1 \circ \psi_1 \circ \varphi_2 \circ \psi_2 \circ \cdots \circ \varphi_L \circ \psi_L$$

- Convenient to use orthogonal weights. $\|P\| = 1$, $\|Q\| = 1$, easier

$$\varphi_\ell(x) = x - h_1 P^T \sigma(Px + p) \quad \text{contractive}$$

$$\psi_\ell(x) = x + h_2 Q^T \sigma(Qx + q) \quad \text{expansive}$$

$$\sigma(x) = \max\left\{x, \frac{x}{2}\right\}, \quad P^T P = I, \quad Q^T Q = I.$$

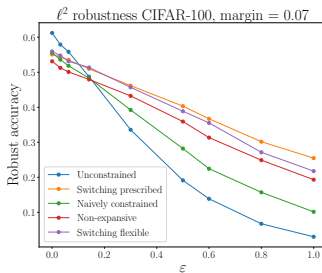
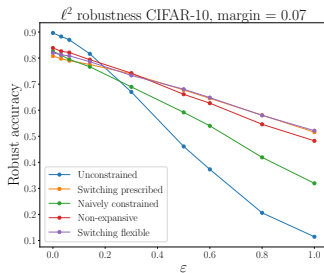
Robust classification of CIFAR10 and CIFAR100

$$\varphi_{\ell}(x) = x - h_1 P^T \sigma(Px + p) \quad \text{contractive}$$

$$\psi_{\ell}(x) = x + h_2 Q^T \sigma(Qx + q) \quad \text{expansive}$$

$$\sigma(x) = \max\left\{x, \frac{x}{2}\right\}, \quad P^T P = I, \quad Q^T Q = I.$$

Using orthogonal convolutional NNs. by Wang et al., 2020. Adversarial examples using Foolbox.

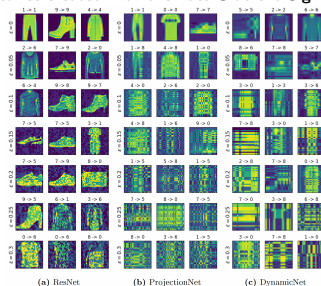


EC, Murari, Owren, Schönlieb and Sherry, *Dynamical systems based neural networks*, 2023, SISC

Goals:

- Study of adversarial robustness.

Adversaries created with the Fast Gradient Sign Method



- C. Balestrand Klemetsen, EC, *Detecting artificially generated data*, in preparation.
- C. Balestrand Klemetsen, *Neural Networks on Low-Rank and Stiefel Manifolds*, Master Thesis.

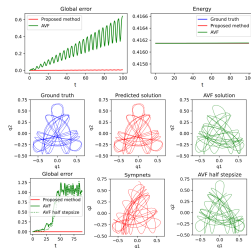
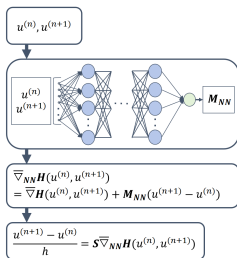
Universal Energy Preserving resp. Dissipative Discrete Gradient Integrators (REMODEL)

- Discrete gradients are a simple approach for numerical discretization of $\dot{u} = S \nabla H(u)$ with S skew-symmetric (energy preserving) or S negative definite (energy dissipative).
- A discrete gradient $\bar{\nabla} H(u^n, u^{n+1})$ satisfies $\bar{\nabla} H(u, u) = \nabla H(u)$ and

$$(u^{n+1} - u^n)^T \bar{\nabla} H(u^n, u^{n+1}) = H(u^{n+1}) - H(u^n)$$

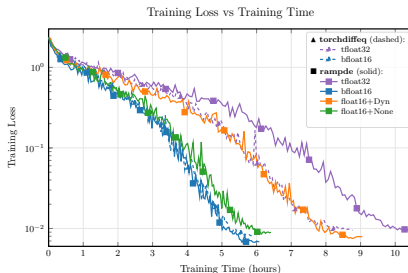
- Characterization of all discrete gradients:

$$\bar{\nabla} H(u^n, u^{n+1}) = \bar{\nabla}_0 H(u^n, u^{n+1}) + M(u^n, u^{n+1})(u^{n+1} - u^n)$$



- UEPI: Universal Energy-Behavior-Preserving Integrators for Energy Conservative/Dissipative Differential Equations, E. Celledoni, B. Owren, C. Shen, B. Xu, T. Yaguchi, NeurIPS 2025.

- Mixed precision solvers and backpropagation
- Dynamic adjoint scaling
- Analysis of roundoff propagation of ResNets (and Neural ODEs)
- Implementation rampde (robust automatic mixed precision for differential equations)

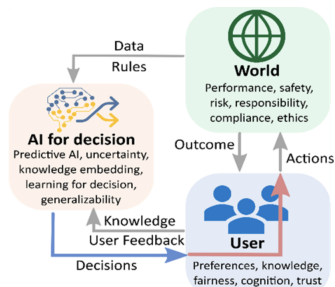


- E Celledoni, B Owren, L Ruthotto, T N Yang, *Mixed Precision Training of Neural ODEs*, arXiv:2510.23498.
- <https://github.com/EmoryMLIP/rampde>

AID is a national center for Artificial Intelligence for Decision Making



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For enhanced performance, robustness and reliability in new knowledge and data-driven decision-making processes, it is important to understand the mathematical foundations and statistical properties of AI-based approaches.

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Thank you for listening!